Homework 5

Portfolios

Due, Sunday Feb 10 at 10 pm On canvas

1 Financial Literacy

Answer these questions in 3 steps. (1) give an answer to the question, (2) look over the material assigned for class and find a definition, (3) modify, if need be, your first answer. The goal is not to memorize a given answer but to be sure you can explain the concept to someone. If you can't, then you do not control that concept.

a. Mean-Variance trade-off

Investors usually prefer higher return over lower return and lower risk over higher risk. The demand hence drives the price so that usually less risky asset has lower returns, and consumers need to adjust their portfolios to balance return (mean) with risk (variance).

b. Risk Aversion

A person is said to be risk averse if he prefers outcomes that are more certain and have lower potential payoff to those with more uncertainty but potentially higher payoff.

c. Expected Utility

It is calculated as a weighted average of the utility given each contingency, using the probabilities of the corresponding outcome as weight.

d. Utility of the expected value

It is calculated by first taking the weighted average of the payoffs and then computing the utility of this weighted average

e. Portfolio with Short sales

An investor may borrow an asset and sell it at the current time, and purchase it to return in the future. He may profit from the decline in the price of that borrowed asset. Short sales allow the portfolio to achieve a higher return.

f. Risk

It usually refers to the uncertainty of an outcome and is usually measured by the likelihood

and extend of its deviation from its expected value.

g. Portfolio Separation

The number of portfolios that are needed to produce an optimum allocation is equal to the number of characteristics that investors care about. In particular, if investors only care about return (mean) and risk (minimizing variance), then every investor's portfolio can be computed as a mix of only two dominant portfolios.

2 Thought problems

1. You are working at the Bank of Econ as an investment strategist. As part of your you find two assets that are perfectly negatively correlated.

   (a) What the best pure long strategy you can propose to a highly risk-averse investor?

   Let the two risky assets have expected returns μ_1, μ_2, standard deviations σ_1, σ_2, and correlation $\rho = -1$. The portfolio variance is

   $$\sigma_p^2 = (w_1\sigma_1 - w_2\sigma_2)^2.$$

   Setting $\sigma_p^2 = 0$ yields

   $$w_1 = \frac{\sigma_2}{\sigma_1 + \sigma_2}, \qquad w_2 = \frac{\sigma_1}{\sigma_1 + \sigma_2}.$$

   This long–long portfolio is riskless and has expected return

   $$\mu_p = w_1\mu_1 + w_2\mu_2,$$

   which strictly dominates holding either asset alone.

   (b) Suppose the same investor ask you to devise a 2x leveraged strategy what do you propose?

   Leverage the zero-variance portfolio by borrowing at the risk-free rate r_f and investing 200% in the portfolio:

   $$\tilde{w}_i = 2w_i \quad (i = 1, 2).$$

   The portfolio remains riskless, with expected return

   $$\mu_{2x} = 2\mu_p - r_f.$$

(c) Now you face an investor who is willing to bear risk. He has gone to the Bank of NotEcon and they have suggested a mix of the 1 year Tbill and the market weighted equity portfolio. Can you beat that?

Yes. The Bank of NotEcon combines the T-bill with the market portfolio. Since the zero-variance portfolio delivers $\mu_p > r_f$ with no risk, both the T-bill and the market portfolio are mean–variance dominated. Thus, your strategy strictly improves upon their investment set.

(d) Draw the investment frontier as used by Bank of NotEcon and then draw a new investment frontier based on your discovery.

The Bank of NotEcon frontier is the Capital Market Line:

$$E[R] = r_f + \frac{\mu_M - r_f}{\sigma_M}\sigma.$$

Your discovery introduces a new risk-free asset with return $\mu_p > r_f$, generating a higher and steeper capital market line. The new frontier lies strictly above the old one.

(e) Is this a market equilibrium? If enough people discover this pair of assets what is likely to happen to their prices?

This is not an equilibrium. If many investors discover this pair of assets, demand will drive prices up and expected returns down until

$$\mu_p = r_f,$$

or the perfect negative correlation disappears. The arbitrage opportunity is eliminated in equilibrium.

2. You have just woken up in a binomial world with wealth W. It means that next month your wealth is either $W^1 = (1+r)W + w$ or $W^1 = (1+r)W - w$.

(a) What does it mean in this world to be risk-averse?

Let next-period wealth be

$$W^1 = (1+r)W \pm w$$

(with probabilities, say, 1/2 and 1/2). An investor is *risk-averse* if for any such

mean-preserving risk,

$$\mathbb{E}[U(W^1)] \leq U(\mathbb{E}[W^1]).$$

Equivalently, U is concave:

$$U''(W) < 0.$$

In words: the investor prefers the *certain* wealth level $\mathbb{E}[W^1] = (1+r)W$ to the risky lottery with the same mean.

(b) You can purchase a contract for price p that guarantees your wealth next period will be $W^1 = (1+r)W$. Do all risk-averse investors purchase it?

Without the contract, expected utility is $\mathbb{E}[U((1+r)W \pm w)]$. With the contract, the investor pays p today, which reduces next-period guaranteed wealth to

$$W^1 = (1+r)(W-p).$$

A risk-averse investor buys the contract iff

$$U\big((1+r)(W-p)\big) \;\geq\; \mathbb{E}\big[U\big((1+r)W \pm w\big)\big].$$

Thus, *not necessarily*: if p is too high, even a risk-averse investor will not buy it. Every risk-averse investor buys it only if p is at most their willingness-to-pay (certainty-equivalent premium).

(c) Suppose an investor's utility function is $U(W^1) = (W^1)^{0.5}$ compute the price this investor is willing to pay for the contract.

Assume $\Pr(+w) = \Pr(-w) = 1/2$. The investor is indifferent when

$$U\big((1+r)(W-p)\big) = \frac{1}{2}U\big((1+r)W+w\big) + \frac{1}{2}U\big((1+r)W-w\big).$$

Using $U(x) = \sqrt{x}$:

$$\sqrt{(1+r)(W-p)} = \frac{1}{2}\sqrt{(1+r)W+w} + \frac{1}{2}\sqrt{(1+r)W-w}.$$

Square both sides and solve for p:

$$(1+r)(W-p) = \left(\frac{\sqrt{(1+r)W+w} + \sqrt{(1+r)W-w}}{2}\right)^2,$$

$$p = W - \frac{1}{1+r}\left(\frac{\sqrt{(1+r)W+w}+\sqrt{(1+r)W-w}}{2}\right)^2.$$

This p is the investor's willingness-to-pay (maximum premium).

(d) The binomial world now has two economies, and r is zero. In E1 every other period there is a wealth shock w and in E2 every period there is a wealth shock w/2. Draw the binomial tree over 2 periods for E1 and E2. Why might a risk averse individual prefer E2 over E1? Does you conclusion change if r¿0.

E1: shock w every other period (over 2 periods: shock only in period 2)

$$\begin{array}{c} W \\ \downarrow \\ W \\ \downarrow \\ \begin{cases} W+w \\ W-w \end{cases} \end{array}$$

E2: shock $w/2$ every period (over 2 periods: shocks in both periods)

$$\begin{array}{ccccc} & & W & & \\ & \swarrow & & \searrow & \\ & W+\frac{w}{2} & & W-\frac{w}{2} & \\ & \swarrow \quad \searrow & & \swarrow \quad \searrow & \\ W+w & & W & & W \quad W-w \end{array}$$

Why a risk-averse individual may prefer E2: Both economies have the same possible terminal wealth levels over 2 periods ($W+w$, W, $W-w$), but E2 *resolves risk earlier* and provides smoother paths (a smaller shock each period). With concave utility, earlier/smoother resolution can raise expected utility (less "waiting" under uncertainty, and less severe interim downside). **If $r > 0$:** The qualitative preference for smoother/earlier resolution typically remains, and can be *stronger* if interim wealth affects decisions (e.g., consumption/saving) because earlier losses reduce the base that compounds at rate r. (If utility depends only on terminal wealth and probabilities, then only the terminal distribution matters; but with intermediate utility/consumption, timing matters.)

3. Suppose you are in an economy 12 assets where the market rules are mean variance and there is no riskless asset. As part of your work, compute the mean-variance frontier.

(a) Suppose you find that there is an asset that is never part of any frontier portfolio. What do you deduce is likely to happen to its price. What strategy could you deploy to take advantage of this insight?

If asset j is never held by any efficient (frontier) portfolio, it is *mean–variance dominated*: there exists another portfolio with

$$E[R] \geq E[R_j] \text{ and } Var(R) \leq Var(R_j),$$

with at least one strict inequality. Hence, in equilibrium with mean–variance investors, demand for asset j is weak, so its price is likely to **fall** (raising its expected return) until it becomes competitive / enters the efficient set.

Strategy: Underweight or short asset j and go long a dominating efficient portfolio (or a close substitute on the frontier). As the price of j declines, you profit from the short (and/or from the relative outperformance of the efficient substitute).

(b) Now suppose there is news that one of your assets is going to have an important positive bump in its returns. What do you deduce is likely to happen to its price. What strategy could you deploy to take advantage of this insight?

A positive revision in expected return for asset k (higher μ_k) increases its desirability in mean–variance optimization, shifting the frontier outward. Thus its price is likely to **rise immediately** (expected return falls via higher price) until the new equilibrium expected return reflects the news.

Strategy: Buy asset k (or increase its weight relative to the current frontier portfolio) as soon as you learn the news. If you can express a relative view, go long k and short a portfolio with similar risk exposure to isolate the idiosyncratic repricing of k.

(c) Now suppose there is news that one of your assets is going to have an important change in its volatility. What do you deduce is likely to happen to its price? What strategy could you deploy to take advantage of this insight?

Let asset m experience a volatility change $\sigma_m \mapsto \sigma'_m$.

- If volatility *increases* ($\sigma'_m > \sigma_m$), the asset becomes less attractive (higher risk per unit of mean), so its price is likely to **fall** (raising expected return) until investors are compensated for the higher risk.
- If volatility *decreases* ($\sigma'_m < \sigma_m$), the asset becomes more attractive, so its price is likely to **rise** (lowering expected return) until equilibrium.

Strategy: If volatility increases, reduce exposure / short asset m and rotate

toward efficient substitutes; if volatility decreases, buy asset m (or tilt toward it) ahead of the repricing. In either case, a relative trade is: long the asset that becomes *more* mean–variance efficient and short a nearby substitute that becomes relatively worse.

4. In many poor economies, farmers plant multiple crops and have fields in different ecological zones of their village.

   (a) Give a mean-variance explanation for this pattern

   Let farm income be a portfolio of crop/plot returns:

$$Y = \sum_{i=1}^{N} w_i R_i, \qquad \sum_{i=1}^{N} w_i = 1, \quad w_i \geq 0,$$

   where R_i is the stochastic return (profit) from crop–zone activity i. Then

$$\mathbb{E}[Y] = \sum_{i=1}^{N} w_i \mu_i, \qquad \mathrm{Var}(Y) = \sum_{i=1}^{N} \sum_{j=1}^{N} w_i w_j \sigma_{ij}.$$

   If rainfall, pests, and soil shocks differ across crops and ecological zones, then covariances σ_{ij} are low (or negative) across some pairs. A risk-averse farmer (concave utility) values lower variance, so diversification across crops/zones reduces $\mathrm{Var}(Y)$ for a given $\mathbb{E}[Y]$. Thus, multi-cropping and spatial diversification are a form of *self-insurance*.

   (b) Give a technological explanation for why this pattern tends to disappear as economies get richer.

   As economies get richer, technologies relax constraints that make returns highly risky and idiosyncratic:

   - *Risk-reducing inputs:* irrigation, improved seeds, fertilizer, pesticides, drainage, mechanization. These reduce σ_i^2 (variance of each activity's returns) and/or raise μ_i.
   - *Better information and control:* weather forecasts, extension services, agronomic knowledge. This reduces unanticipated shocks and improves input timing, lowering effective variance.
   - *Scale and specialization:* machinery and fixed costs create gains to specializing in fewer crops, and improved control makes specialization less risky.

When σ_i^2 falls and expected return gains from specialization rise, the optimal portfolio becomes less diversified.

(c) Give a market-based explanation for why this pattern tends to disappear as economies get richer. What sets of markets matter? be sure to explain your answer. Diversification is often a substitute for missing or imperfect markets. As richer economies develop, farmers can *offload risk* and *smooth consumption* through markets, reducing the need to diversify on-farm.
Key markets:

- *Insurance markets (formal or informal):* crop insurance, rainfall/index insurance. These reduce the need to lower $\mathrm{Var}(Y)$ via diversification because downside risk is partially insured.
- *Credit/savings markets:* access to loans, savings accounts, and microfinance allows intertemporal smoothing, so farmers choose higher-μ (even if higher-σ) specialized technologies and smooth consumption via borrowing/saving.
- *Labor markets:* seasonal migration and local wage work allow households to diversify *income sources* off-farm; farm income risk can be offset by wage income, reducing the need for crop/zonal diversification.
- *Output/input markets and trade integration:* thicker markets and better price transmission reduce marketing risk and local price volatility; storage and forward contracting can reduce price risk. More reliable input supply chains also reduce production risk.

In summary, as insurance + finance + labor + trade/input/output markets deepen, households can achieve risk-sharing and smoothing *off the farm*, so the privately optimal choice shifts toward specialization.

3 Mean Variance trade-off (single asset problem)

4 Efficiency frontier (Portfolio problem)

The solution is omitted because the answers to these two problems depend on your choice of assets in the financial journal, and the required code is similar to that used in Problem 4, including the mathematical details for the efficient frontier.

5 Personal finance

You are a financial advisor advising a client about what investment portfolios to choose. Using the linked CSV file that you can import into Excel or a statistical package, propose an optimal portfolio in each of the cases below. In each case, your only investment choices are 3-months T-Bills, and three stock funds (NASDAQ, DOW, and Russel 2000).

How to use this solution: This solution is written for a closely related specification (e.g., a different investment horizon), but the underlying derivation and computational steps are identical to those required for the current problem. I recommend first working through the problem on your own before consulting this write-up. You can then use the solution as a consistency check and apply the same procedure to the parameters given in the actual problem set.

A. Compute the mean weekly return and the variance-covariance matrix for every five years 2000-2004, 2005-2009, 2010-2014, 2015-2019, and 2020-2024 and for the incomplete 2025-2030 .as well as for the whole period.

I assume that the T-bill yield is annualized. And to make the numbers comparable, I also annualized the weekly return of stocks using the following formula:

$$\text{Weekly Return} = (1 + \text{Annualized Return})^{1/52} - 1$$

Also, the returns are in percentage

```
[1] "2000 to 2004"
   r_TBill   r_Nasdaq  r_Russell      r_Dow
 0.0004880 -0.0013653  0.0012642  0.0001015
             r_TBill   r_Nasdaq  r_Russell      r_Dow
r_TBill    1.176e-07 -1.470e-06 -5.055e-07 -2.540e-07
r_Nasdaq  -1.470e-06  2.136e-03  1.234e-03  4.739e-04
r_Russell -5.055e-07  1.234e-03  1.041e-03  4.416e-04
r_Dow     -2.540e-07  4.739e-04  4.416e-04  5.957e-04

[1] "2005 to 2009"
  r_TBill  r_Nasdaq r_Russell     r_Dow
0.0005092 0.0008123 0.0007261 0.0003518
            r_TBill  r_Nasdaq r_Russell     r_Dow
r_TBill   1.225e-07 1.304e-07 1.956e-07 3.733e-07
r_Nasdaq  1.304e-07 9.723e-04 1.052e-03 5.283e-04
r_Russell 1.956e-07 1.052e-03 1.355e-03 7.206e-04
r_Dow     3.733e-07 5.283e-04 7.206e-04 7.483e-04

[1] "2010 to 2014"
  r_TBill  r_Nasdaq r_Russell     r_Dow
1.196e-05 3.181e-03 2.860e-03 2.246e-03
            r_TBill  r_Nasdaq r_Russell     r_Dow
r_TBill   8.035e-11 1.877e-09 1.084e-08 2.565e-09
r_Nasdaq  1.877e-09 5.196e-04 5.511e-04 2.786e-04
r_Russell 1.084e-08 5.511e-04 7.052e-04 3.435e-04
r_Dow     2.565e-09 2.786e-04 3.435e-04 3.555e-04

[1] "2015 to 2019"
  r_TBill  r_Nasdaq r_Russell     r_Dow
0.0002028 0.0027686 0.0015457 0.0020471
             r_TBill  r_Nasdaq  r_Russell     r_Dow
r_TBill    2.561e-08 1.113e-08 -5.581e-08 3.123e-08
r_Nasdaq   1.113e-08 5.534e-04  4.323e-04 2.073e-04
r_Russell -5.581e-08 4.323e-04  5.282e-04 2.557e-04
r_Dow      3.123e-08 2.073e-04  2.557e-04 2.946e-04
```

```
[1] "2015 to 2019"
  r_TBill  r_Nasdaq r_Russell     r_Dow
0.0002028 0.0027686 0.0015457 0.0020471
             r_TBill  r_Nasdaq  r_Russell     r_Dow
r_TBill    2.561e-08 1.113e-08 -5.581e-08 3.123e-08
r_Nasdaq   1.113e-08 5.534e-04  4.323e-04 2.073e-04
r_Russell -5.581e-08 4.323e-04  5.282e-04 2.557e-04
r_Dow      3.123e-08 2.073e-04  2.557e-04 2.946e-04

[1] "2020 to 2024"
  r_TBill  r_Nasdaq r_Russell     r_Dow
0.0004668 0.0035621 0.0019112 0.0017773
            r_TBill  r_Nasdaq r_Russell     r_Dow
r_TBill   1.862e-07 4.059e-07 9.013e-08 1.793e-07
r_Nasdaq  4.059e-07 1.062e-03 8.911e-04 4.485e-04
r_Russell 9.013e-08 8.911e-04 1.370e-03 7.302e-04
r_Dow     1.793e-07 4.485e-04 7.302e-04 6.443e-04

[1] "Whole period: 2000 to 2030"
  r_TBill  r_Nasdaq r_Russell     r_Dow
0.0003354 0.0017902 0.0016612 0.0013045
             r_TBill   r_Nasdaq  r_Russell      r_Dow
r_TBill    1.284e-07 -3.831e-07 -1.656e-07 -7.175e-08
r_Nasdaq  -3.831e-07  1.048e-03  8.299e-04  3.875e-04
r_Russell -1.656e-07  8.299e-04  9.966e-04  4.968e-04
r_Dow     -7.175e-08  3.875e-04  4.968e-04  5.265e-04
```

As can be seen from the covariance table above, the variance of T-Bill is much smaller than the stock, so does the covariance between T-Bill and stock. So, it is OK to treat the T-Bill as risk-free. However, in the following calculation, we use the covariance matrix above instead.

B.Compute the efficient frontier for the data over the whole period.

Deonote w as the weights of assets in the portfolio, and μ as the vector of corresponding returns, and $\sum$ as the covairnace matrix. The problem can be thought of as the following constrained optimization problem:

$$\min \omega' \sum \omega$$

$$s.t. \omega\mu \geq r_t$$

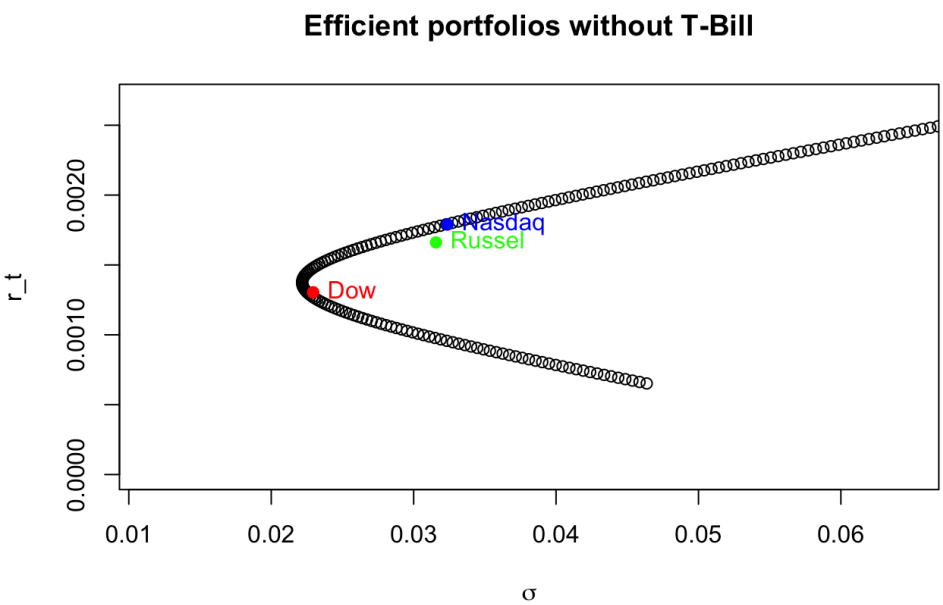
$$\omega'\mathbf{1} = 1$$

It is equivalent to optimize the following Lagrangian method(Taking the first order condition):

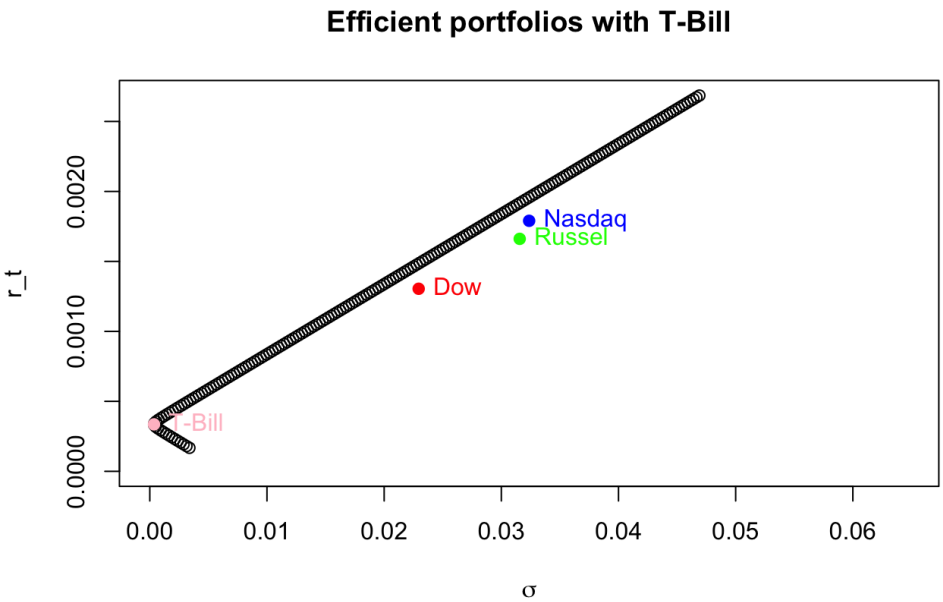
$$\min_{\omega,\lambda_1,\lambda_2} \omega' \sum \omega + \lambda_1(\omega'\mu - r_t) + \lambda_2(\omega'\mathbf{1} = 1)$$

$$\begin{aligned} FOC = & \frac{\partial L(\omega,\lambda_1,\lambda_2)}{\partial \omega} = 2\sum \omega + \lambda_1\mu + \lambda_2\mathbf{1} \\ & \frac{\partial L(\omega,\lambda_1,\lambda_2)}{\partial \lambda_1} = \omega'\mu - r_t = 0 \\ & \frac{\partial L(\omega,\lambda_1,\lambda_2)}{\partial \lambda_2} = x'\mathbf{1} - 1 = 0 \end{aligned}$$

In the matrix from, it is $\begin{pmatrix} 2\sum & \mu & \mathbf{1} \\ \mu' & 0 & 0 \\ \mathbf{1}' & 0 & 0 \end{pmatrix} \begin{pmatrix} \omega \\ \lambda_1 \\ \lambda_2 \end{pmatrix} = \begin{pmatrix} 0 \\ r_t \\ 1 \end{pmatrix}$



By iterating over different values of target rate, we can describe the efficient frontier. Above is a scatter plot of the efficient portfolios without consideration of the T-Bill.



Above is a scatter plot of the efficient portfolios with consideration of the T-Bill.

C. Now assume you have a client with a target return of 5% per year. What is the efficient portfolio given that return target?

A annual return of 5% converted to the weekly base is $(1 + 5\%)^{1/52} - 1 = 0.0009387$. We plug in the target rate in the above problem, and get:

T-Bill	Nasdaq	Russel	Dow
0.49211	0.25763	-0.03934	0.28960

For a return of 2% we get:

T-Bill	Nasdaq	Russel	Dow
0.961799	0.019827	-0.003359	0.021733

D. Your client arrives on January 15, 2005, with a five-year horizon. What is the efficient portfolio given the data 2000-2004 given his target return? What is its performance given the realized returns 2005-2019

We then compute the return and covariance matrix using data from 2000-2004, and get the efficient portfolio for 5%,2%target rate respectively:

T-Bill	Nasdaq	Russel	Dow
1.04519	-0.15260	0.18014	-0.07274

T-Bill	Nasdaq	Russel	Dow
0.98881	0.03640	-0.04244	0.01723

Assuming that the client holds this portfolio for 5-years. For the portfolio with target rate of 5%, he will get a annual return of $(1 + \omega'\mu_{05-09})^{52} - 1 = 2.71\%$; For that of 2%, he will get an annual return of 2.68%

E. Your client arrives on January 1, 2010, with a five-year horizon. What is the portfolio you propose given the data 2000-09? What is its performance given the realized returns from 2010-2014?

We then compute the return and covariance matrix using data from 2000-2009, and get the efficient portfolio for 5%, 2% target rate respectively:

T-Bill	Nasdaq	Russel	Dow
1.0939	-0.2778	0.3586	-0.1747

T-Bill	Nasdaq	Russel	Dow
0.97444	0.07453	-0.09557	0.04660

Assuming that the client holds this portfolio for 5-years. For the portfolio with target rate of 5%, he will get a annual return of $(1 + \omega'\mu_{10-14})^{52} - 1 = -1.23\%$; For that of 2%, he will get an annual return of 0.417% .

F. Would your client have been better off staying with the weights from question D?
Using the portfolio from D, the client will instead get -0.625% if his target rate is 5%; and get 0.234% if his target rate is 2%. Hence, we conclude that he will be better if staying with the porfolio from D)

G. Your client arrives on January 1, 2015, with a five-year horizon. What is the portfolio you propose, given the data from 2010-2014? What is its performance given the realized returns for 2015-2020?
Suppose we use the data from 10-14 and hold the portfolio during 15-20. The efficient portfolio for target rate of 5% and 2% are respectively:

T-Bill	Nasdaq	Russel	Dow
0.64532	0.18073	-0.05638	0.23033

T-Bill	Nasdaq	Russel	Dow
0.85885	0.07221	-0.02267	0.09160

For the portfolio with target rate of 5%, he will get a annual return of 5.42%; For that of 2%, he will get an annual return of 2.78%.

Code:

```
## Load data and packages
library(dplyr)
library(tidyverse)
library(lubridate)
library(ggplot2)
data <- read.csv("/Users/gaoyue/Desktop/HW5-data.csv",
                 skip = 1,
                 col.names = c("date", "TBill", "Nasdaq", "Russell", "Dow")) %>%
  mutate(date = ymd(date)) %>%                  # Convert date column to date format
  filter(!is.na(date)) %>%                      # Eliminate rows with NA dates
  mutate(
    r_Nasdaq = (Nasdaq - lag(Nasdaq)) / lag(Nasdaq),       # Calculate returns for Nasdaq
    r_Russell = (Russell - lag(Russell)) / lag(Russell),   # Returns for Russell
    r_Dow = (Dow - lag(Dow)) / lag(Dow),                   # Returns for Dow
    r_TBill = (1 + TBill / 100)^(1 / 52) - 1               # Weekly returns for TBill
  )

# Set global display options
options(digits = 4)
```

```r
#### A)
for (yr in seq(2000,2025,5)) {
  start_date <- paste(yr,"-01-31",sep = "")
end_date <-paste(yr+5,"-01-31",sep = "")
subdata<-data%>%filter(date>=start_date, date<end_date)%>%
  select(r_TBill,r_Nasdaq,r_Russell,r_Dow)
if(yr<2025){
  print(paste0(yr," to ",yr+4))
}
else{
  print("2025 to 2030")
}
print(colMeans(subdata,na.rm = T))
print(cov(subdata,use = "complete.obs"))
#print(cov(subdata%>%select(r_Nasdaq,r_Russell,r_Dow),use = "complete.obs"))
cat("\n")
if(yr>=2020){
  print("Whole period: 2000 to 2030")
  print(colMeans(data%>%select(r_TBill,r_Nasdaq,r_Russell,r_Dow),na.rm = T))
  print(cov(data%>%select(r_TBill,r_Nasdaq,r_Russell,r_Dow),use = "complete.obs"))
}
}
```

```r
#### B)
mu.vec = colMeans(data%>%select(r_Nasdaq,r_Russell,r_Dow),na.rm = T)
sigma.mat <- matrix(cov(data%>%select(r_Nasdaq,r_Russell,r_Dow),use = "complete.obs"),nrow=3)
dimnames(sigma.mat)<-list(names(mu.vec),names(mu.vec))
top.mat = cbind(2*sigma.mat, mu.vec, rep(1, 3))
mid.vec = c(mu.vec, 0, 0)
bot.vec = c(rep(1, 3), 0, 0)
Ax.mat = rbind(top.mat, mid.vec, bot.vec)
meanVarP <-function(r_t){
  #A function that gives the min-variance portfolio for given target return
  bmsft.vec = c(rep(0, 3), r_t, 1)
  z.mat = solve(Ax.mat)%*%bmsft.vec
  x.vec = z.mat[1:3,]
  mu.p = as.numeric(crossprod(x.vec, mu.vec))
  sigma.p = sqrt(as.numeric(t(x.vec)%*%sigma.mat%*%x.vec))
  return(list(w=x.vec,mu=mu.p,sigma=sigma.p))
}
r_ts <- seq(min(mu.vec)/2,max(mu.vec)*1.5,(max(mu.vec)*1.5-min(mu.vec)/2)/200)
sigma.ps<-c()
for(r_t in r_ts){
  sigma.ps<-c(sigma.ps,meanVarP(r_t)$sigma)
}
plot(sigma.ps,r_ts, xlab = expression(sigma),
     ylab = expression(r_t),xlim=c(sqrt(min(diag(sigma.mat)))/2,sqrt(max(sigma.mat))*2),
     ylim=c(0,max(mu.vec)*1.5),
     main="Efficient portfolios without T-Bill")
points(sqrt(sigma.mat[1,1]),mu.vec[1], col = "blue", pch = 19)
text(sqrt(sigma.mat[1,1]),mu.vec[1], labels = 'Nasdaq', pos = 4, col = "blue")
points(sqrt(sigma.mat[2,2]),mu.vec[2], col = "green", pch = 19)
text(sqrt(sigma.mat[2,2]),mu.vec[2], labels = 'Russel', pos = 4, col = "green")
points(sqrt(sigma.mat[3,3]),mu.vec[3], col = "red", pch = 19)
text(sqrt(sigma.mat[3,3]),mu.vec[3], labels = 'Dow', pos = 4, col = "red")
```

```r
#### B)
mu.vec = colMeans(data%>%select(r_TBill,r_Nasdaq,r_Russell,r_Dow),na.rm = T)
sigma.mat <- matrix(cov(data%>%select(r_TBill,r_Nasdaq,r_Russell,r_Dow),use = "complete.obs"),nrow=4)
dimnames(sigma.mat)<-list(names(mu.vec),names(mu.vec))
top.mat = cbind(2*sigma.mat, mu.vec, rep(1, 4))
mid.vec = c(mu.vec, 0, 0)
bot.vec = c(rep(1, 4), 0, 0)
Ax.mat = rbind(top.mat, mid.vec, bot.vec)
meanVarP <-function(r_t){
  #A function that gives the min-variance portfolio for given target return
  bmsft.vec = c(rep(0, 4), r_t, 1)
  z.mat = solve(Ax.mat)%*%bmsft.vec
  x.vec = z.mat[1:4,]
  mu.p = as.numeric(crossprod(x.vec, mu.vec))
  sigma.p = sqrt(as.numeric(t(x.vec)%*%sigma.mat%*%x.vec))
  return(list(w=x.vec,mu=mu.p,sigma=sigma.p))
}
r_ts <- seq(min(mu.vec)/2,max(mu.vec)*1.5,(max(mu.vec)*1.5-min(mu.vec)/2)/200)
sigma.ps<-c()
for(r_t in r_ts){
  sigma.ps<-c(sigma.ps,meanVarP(r_t)$sigma)
}
plot(sigma.ps,r_ts, xlab = expression(sigma),
     ylab =expression(r_t),xlim=c(0,sqrt(max(sigma.mat))*2),
     ylim=c(0,max(mu.vec)*1.5),
     main="Efficient portfolios with T-Bill"
)
points(sqrt(sigma.mat[1,1]),mu.vec[1], col = "pink", pch = 19)
text(sqrt(sigma.mat[1,1]),mu.vec[1], labels = 'T-Bill', pos = 4, col = "pink")
points(sqrt(sigma.mat[2,2]),mu.vec[2], col = "blue", pch = 19)
text(sqrt(sigma.mat[2,2]),mu.vec[2], labels = 'Nasdaq', pos = 4, col = "blue")
points(sqrt(sigma.mat[3,3]),mu.vec[3], col = "green", pch = 19)
text(sqrt(sigma.mat[3,3]),mu.vec[3], labels = 'Russel', pos = 4, col = "green")
points(sqrt(sigma.mat[4,4]),mu.vec[4], col = "red", pch = 19)
text(sqrt(sigma.mat[4,4]),mu.vec[4], labels = 'Dow', pos = 4, col = "red")
```

```r
#### C)
r_t5 <- 1.05^(1/52)-1
w_5<-meanVarP(r_t5)$w
names(w_5)<-c("T-Bill","Nasdaq","Russel","Dow")
w_5
r_t2 <- 1.02^(1/52)-1
w_2<-meanVarP(r_t2)$w
names(w_2)<-c("T-Bill","Nasdaq","Russel","Dow")
w_2
```

```
#### D)
start_date <- paste(2000,"-01-31",sep = "")
end_date <-paste(2005,"-01-31",sep = "")
mu.vec = colMeans(data%>%filter(date>=start_date,date<end_date)%>%
                    select(r_TBill,r_Nasdaq,r_Russell,r_Dow),na.rm = T)
sigma.mat <- matrix(cov(data%>%select(r_TBill,r_Nasdaq,r_Russell,r_Dow),use = "complete.obs"),nrow=4)
dimnames(sigma.mat)<-list(names(mu.vec),names(mu.vec))
top.mat = cbind(2*sigma.mat, mu.vec, rep(1, 4))
mid.vec = c(mu.vec, 0, 0)
bot.vec = c(rep(1, 4), 0, 0)
Ax.mat = rbind(top.mat, mid.vec, bot.vec)
meanVarP04 <-function(r_t){
  #A function that gives the min-variance portfolio for given target return
  bmsft.vec = c(rep(0, 4), r_t, 1)
  z.mat = solve(Ax.mat)%*%bmsft.vec
  x.vec = z.mat[1:4,]
  mu.p = as.numeric(crossprod(x.vec, mu.vec))
  sigma.p = sqrt(as.numeric(t(x.vec)%*%sigma.mat%*%x.vec))
  return(list(w=x.vec,mu=mu.p,sigma=sigma.p))
}
w_D5<-meanVarP04(r_t5)$w
names(w_D5)<-c("T-Bill","Nasdaq","Russel","Dow")
w_D5
w_D2<-meanVarP04(r_t2)$w
names(w_D2)<-c("T-Bill","Nasdaq","Russel","Dow")
w_D2
start_date <- paste(2005,"-01-31",sep = "")
end_date <-paste(2010,"-01-31",sep = "")
mu_09<-colMeans(data%>%filter(date>=start_date, date<end_date)%>%
                  select(r_TBill,r_Nasdaq,r_Russell,r_Dow),na.rm = T)
r_pD5 <- t(w_D5)%*%mu_09
return_D5<-(1+r_pD5)^(52)-1
r_pD2 <- t(w_D2)%*%mu_09
return_D2<-(1+r_pD2)^(52)-1
```

```
#### E)
start_date <- paste(2000,"-01-31",sep = "")
end_date <-paste(2010,"-01-31",sep = "")
mu.vec = colMeans(data%>%filter(date>=start_date,date<end_date)%>%
                    select(r_TBill,r_Nasdaq,r_Russell,r_Dow),na.rm = T)
sigma.mat <- matrix(cov(data%>%select(r_TBill,r_Nasdaq,r_Russell,r_Dow),use = "complete.obs"),nrow=4)
dimnames(sigma.mat)<-list(names(mu.vec),names(mu.vec))
top.mat = cbind(2*sigma.mat, mu.vec, rep(1, 4))
mid.vec = c(mu.vec, 0, 0)
bot.vec = c(rep(1, 4), 0, 0)
Ax.mat = rbind(top.mat, mid.vec, bot.vec)
meanVarP09 <-function(r_t){
  #A function that gives the min-variance portfolio for given target return
  bmsft.vec = c(rep(0, 4), r_t, 1)
  z.mat = solve(Ax.mat)%*%bmsft.vec
  x.vec = z.mat[1:4,]
  mu.p = as.numeric(crossprod(x.vec, mu.vec))
  sigma.p = sqrt(as.numeric(t(x.vec)%*%sigma.mat%*%x.vec))
  return(list(w=x.vec,mu=mu.p,sigma=sigma.p))
}
w_E5<-meanVarP09(r_t5)$w
names(w_E5)<-c("T-Bill","Nasdaq","Russel","Dow")
w_E5
w_E2<-meanVarP09(r_t2)$w
names(w_E2)<-c("T-Bill","Nasdaq","Russel","Dow")
w_E2
start_date <- paste(2010,"-01-31",sep = "")
end_date <-paste(2015,"-01-31",sep = "")
mu_15<-colMeans(data%>%filter(date>=start_date, date<end_date)%>%
                  select(r_TBill,r_Nasdaq,r_Russell,r_Dow),na.rm = T)
r_pE5 <- t(w_E5)%*%mu_15
return_E5<-(1+r_pE5)^(52)-1
r_pE2 <- t(w_E2)%*%mu_15
return_E2<-(1+r_pE2)^(52)-1
```

```
#### F)
r_pF5 <- t(w_D5)%*%mu_15
return_F5<-(1+r_pF5)^(52)-1
r_pF2 <- t(w_D2)%*%mu_15
return_F2<-(1+r_pF2)^(52)-1
```

```
#### G)
start_date <- paste(2010,"-01-31",sep = "")
end_date <-paste(2015,"-01-31",sep = "")
mu.vec = colMeans(data%>%filter(date>=start_date,date<end_date)%>%
                  select(r_TBill,r_Nasdaq,r_Russell,r_Dow),na.rm = T)
sigma.mat <- matrix(cov(data%>%select(r_TBill,r_Nasdaq,r_Russell,r_Dow),use = "complete.obs"),nrow=4)
dimnames(sigma.mat)<-list(names(mu.vec),names(mu.vec))
top.mat = cbind(2*sigma.mat, mu.vec, rep(1, 4))
mid.vec = c(mu.vec, 0, 0)
bot.vec = c(rep(1, 4), 0, 0)
Ax.mat = rbind(top.mat, mid.vec, bot.vec)
meanVarP14 <-function(r_t){
  #A function that gives the min-variance portfolio for given target return
  bmsft.vec = c(rep(0, 4), r_t, 1)
  z.mat = solve(Ax.mat)%*%bmsft.vec
  x.vec = z.mat[1:4,]
  mu.p = as.numeric(crossprod(x.vec, mu.vec))
  sigma.p = sqrt(as.numeric(t(x.vec)%*%sigma.mat%*%x.vec))
  return(list(w=x.vec,mu=mu.p,sigma=sigma.p))
}
w_G5<-meanVarP09(r_t5)$w
names(w_G5)<-c("T-Bill","Nasdaq","Russel","Dow")
w_G5

w_G2<-meanVarP09(r_t2)$w
names(w_G2)<-c("T-Bill","Nasdaq","Russel","Dow")
w_G2
start_date <- paste(2015,"-01-31",sep = "")
end_date <-paste(2020,"-01-31",sep = "")
mu_20<-colMeans(data%>%filter(date>=start_date, date<end_date)%>%
                select(r_TBill,r_Nasdaq,r_Russell,r_Dow),na.rm = T)
r_pG5 <- t(w_G5)%*%mu_20
return_G5<-(1+r_pG5)^(52)-1
r_pG2 <- t(w_G2)%*%mu_20
return_G2<-(1+r_pG2)^(52)-1
```